

LAWBOT – ML-BASED CHATBOTS FOR ASSISTANCE ON LEGAL MATTERS IN INDIA

Group ID: 17

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PROBLEM DEFINITION

The problem arises from the fact that access to legal assistance in India is often costly, making it unaffordable for a large section of society. Many individuals are left helpless because lawyers sometimes refuse to take up cases due to pressure or disinterest, while in other cases, corruption in the form of bribes leads to intentional underperformance. As a result, people facing legal troubles often find themselves confused and without proper guidance. Moreover, there is no reliable system in place to help individuals present their cases in a structured and legally sound manner on their own.

- **LABOUR AND EMPLOYMENT LAW**
 - Salary not paid
 - Unlawful termination
 - Workplace harassment
 - Non-issuance of offer letters / contracts
 - Provident fund issues
- **TENANCY / PROPERTY LAW**
 - Landlord not returning deposit
 - Illegal eviction
 - Rent hike without notice
 - Denial of basic services (electricity/water)

LITERATURE SURVEY

Paper Title & Year	Authors	Advantages	Disadvantages	Methodology Used	Key Analysis & Findings	Info Already Used	ML Model Scope & Use
CiteCaseLAW: Citation Worthiness Detection in Caselaw (2023)	Mann Khatri, Reshma Sheik, Pritish Wadhwa, Gitansh Satija, Yaman Kumar, Rajiv Ratn Shah, P. Kumaraguru	Large-scale dataset (178M sentences); High F1-score (88.3%) with LEGAL-BERT+PU; Strong citation detection support	Dataset limited to U.S. caselaw; High computational cost (36 hrs/epoch); Limited gold-standard validation (1k examples)	Created citation-worthiness dataset from Caselaw Access Project; Applied deep learning models (BERT, LEGAL-BERT, PU learning); Used explainable AI (InputXGradient)	LEGAL-BERT+PU outperformed baselines (Precision 84.17, Recall 92.86, F1 88.3); Comparable transfer performance on UNFAIR-ToS and LEDGAR datasets	Concept of citation-aware modeling for legal NLP tasks	ML models (BERT, LEGAL-BERT, CRNN, Transformers); Citation detection and transfer learning in legal domain
Enhancing Public Access to Legal Knowledge in India: Legal Chatbot using Legal-BERT, GPT-2, RAG (2025)	Abhinav Garlapati, Hemanth Koutharapu, Neha Doddi	Context-aware responses; High accuracy (82.5%); User satisfaction (83%); Real-time performance (1.4s response)	Error rate of 12%; Limited coverage to Indian law; Future work needed for multilingual and real-time legal updates	Fine-tuned Legal-BERT on Indian statutes and case law; GPT-2 for human-like response generation; RAG for retrieval of legal documents	Outperformed baselines (BERT, RoBERTa, DistilBERT, Rule/Keyword-based); Accuracy 82.5%, Relevance 84%, Error rate 12%	Legal-BERT already considered in project scope; Similar approach with RAG and GPT-2 for Indian legal chatbot	Legal-BERT for domain understanding; GPT-2 for generation; RAG for retrieval; Applied to Indian legal system

LEGAL DATA COLLECTION MODULE

Collect legal documents (e.g., court judgements, acts, case laws) from authentic sources or user uploads.

DATA PREPROCESSING & STRUCTING

Clean and normalize legal texts, annotate laws, and organize case details for training.

MODEL TRAINING MODULE

Train NLP models (e.g, LawBERT/GPT) on structured legal datasets for classification and advice generation.

QUERY PROCESSING & ADVICE ENGINE

Accept user queries in natural language and generate relevant legal advice, citing laws/precedents.

NONFUNCTIONAL REQUIREMENTS

USABILITY

Interface should be intuitive and simple for non-technical users (e.g., citizens seeking legal help).

MAINTAINIBILITY

System should use modular design easily integrate new laws, case types, or updated models.

SECURITY

Protect sensitive user/legal data using encryption and proper access control.

PORTABILITY

The chatbot should be deployable across platforms (web, mobile, desktop) with minimal setup.

DATASET DESCRIPTION

- **DATASET DESCRIPTION**

- **Data Sources:** Data will be collected from credible and authentic sources such as:
 - **Indian Legal Databases:** Indian Kanoon, Bare Acts, Bar Council portals.
 - **Official Court Websites:** Portals of the Supreme Court of India and various High Courts.
 - **User Uploads:** Integration for processing user-provided legal documents.
 - **Legal Publications:** Digitized books and journals on Indian law.
- **Attributes:** Raw datasets will include unstructured legal text with features such as **Case Title**, **Date**, **Jurisdiction**, **Document Type**, and the **Raw Text** of the document.
- **Derived Attributes:** Using data preprocessing and feature engineering, new attributes such as **Legal Category**, **Summary**, **Problem Statement**, **Actionable Steps**, and **Citations** will be generated and used to enhance the accuracy of the ML model.

PROJECT PLAN

- **Phase 1: Data Collection**
- **Objective:** Gather large-scale legal data specific to Indian contexts.
 - Collect Indian court judgments (Supreme Court, High Courts, Tribunals).
 - Scrape legal platforms (e.g., Indian Kanoon, BareActs, Bar Council portals).
 - Use **OCR tools** (e.g., Tesseract or Google Vision API) for scanned PDFs.
 - Convert unstructured legal documents into machine-readable format (JSON/XML).

PROJECT PLAN

- **Phase 2: Data Annotation & Structuring**
 - **Objective:** Clean, label, and organize legal data for model training.
 - Classify by **law category** (e.g., Tenancy Law, Labour Law).
 - Annotate each case with:
 - **Statutory provisions** (e.g., Section 108 of Transfer of Property Act)
 - **Judgment outcome** (won/lost/settled)
 - **Legal issue, steps advised, and citations**

PROJECT PLAN

- **Phase 3: Dataset Creation & Preprocessing**
 - **Objective:** Prepare high-quality, domain-specific legal training data.
 - Normalize legal terms and eliminate redundant clauses.
 - Extract:
 - **Problem statement** (plain-language)
 - **Actionable legal steps**
 - **Laws and relevant sections**
 - **Judgment citations**

PROJECT PLAN

- **Phase 4: Model Development**
- **Objective:** Train/fine-tune legal NLP models for intelligent legal assistance.
- **Approach:**
 - Use **BERT** or **LawBERT** for legal **retrieval** and classification tasks.
 - Fine-tune **GPT** or GPT-like models to **generate structured legal outputs**.
 - Implement **Retrieval-Augmented Generation (RAG)** to feed retrieved laws/judgments to the generator.
 - Combine citation-aware model (from *CiteCaseLAW*) with language generation (from *IRJMETs 2025*).
- **Key Functions:**
 - Understand user legal queries in natural language.
 - Retrieve relevant laws and judgments from Indian legal corpus.
 - Generate:
 - Legal advice
 - Recommended actions
 - Cited laws and precedents

PROJECT PLAN

- **Phase 5: Evaluation & Validation**

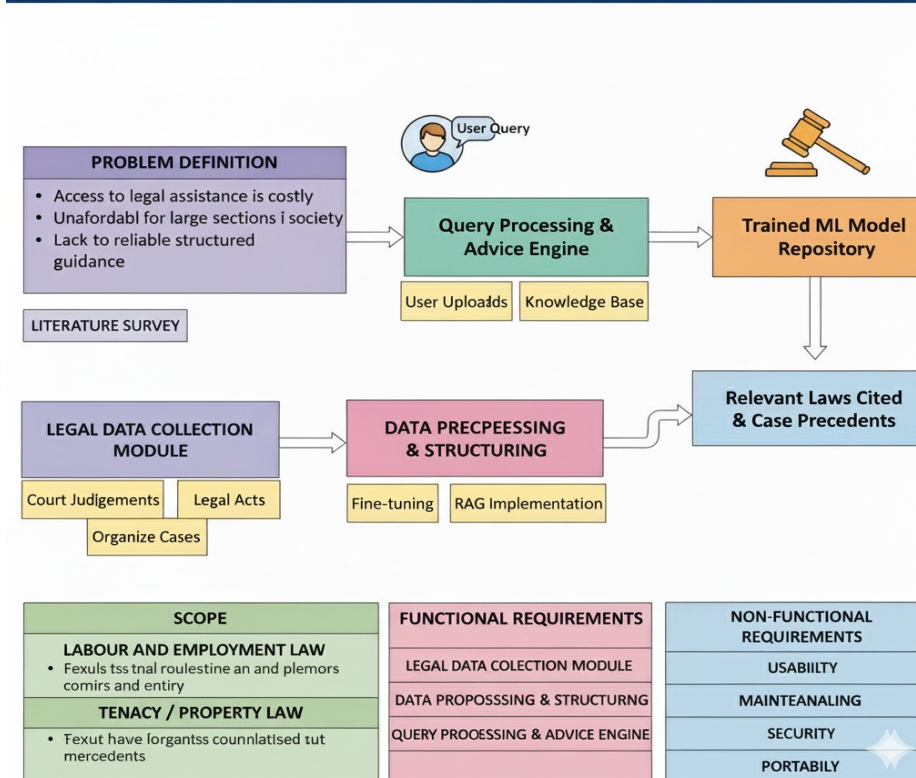
- Validate:
 - Relevance of advice
 - Correctness of law references
 - Suitability of generated arguments
- Measure: Accuracy, F1-score (for classification); BLEU or ROUGE (for generation)

TECHNOLOGIES USED

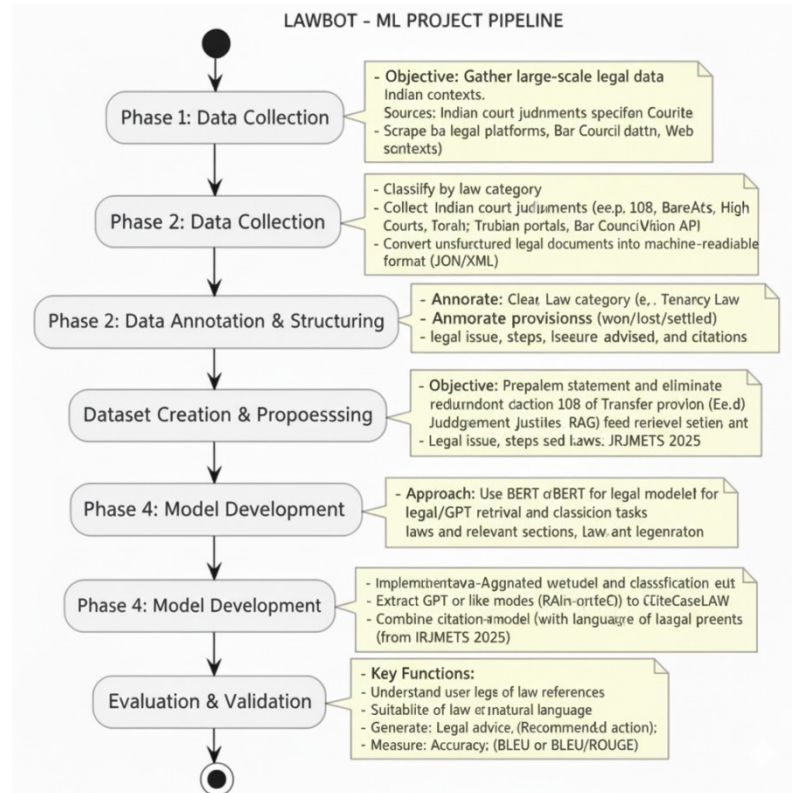
LAWBOT – KEY TECHNOLOGIES & LIBRARIES

Category	Tools / Libraries
Python	
Machine Learning Framework	
File Handling	
Text Preprocessing / ML Ops	
Data Storage / Retrieval	 PostSGGRL (ee.gl., FAISS)
Cloud Platform / Deployment	 

LAWBOT - ML-BASED CHATBOTS FOR LEGAL ASSISTANCE IN INDIA

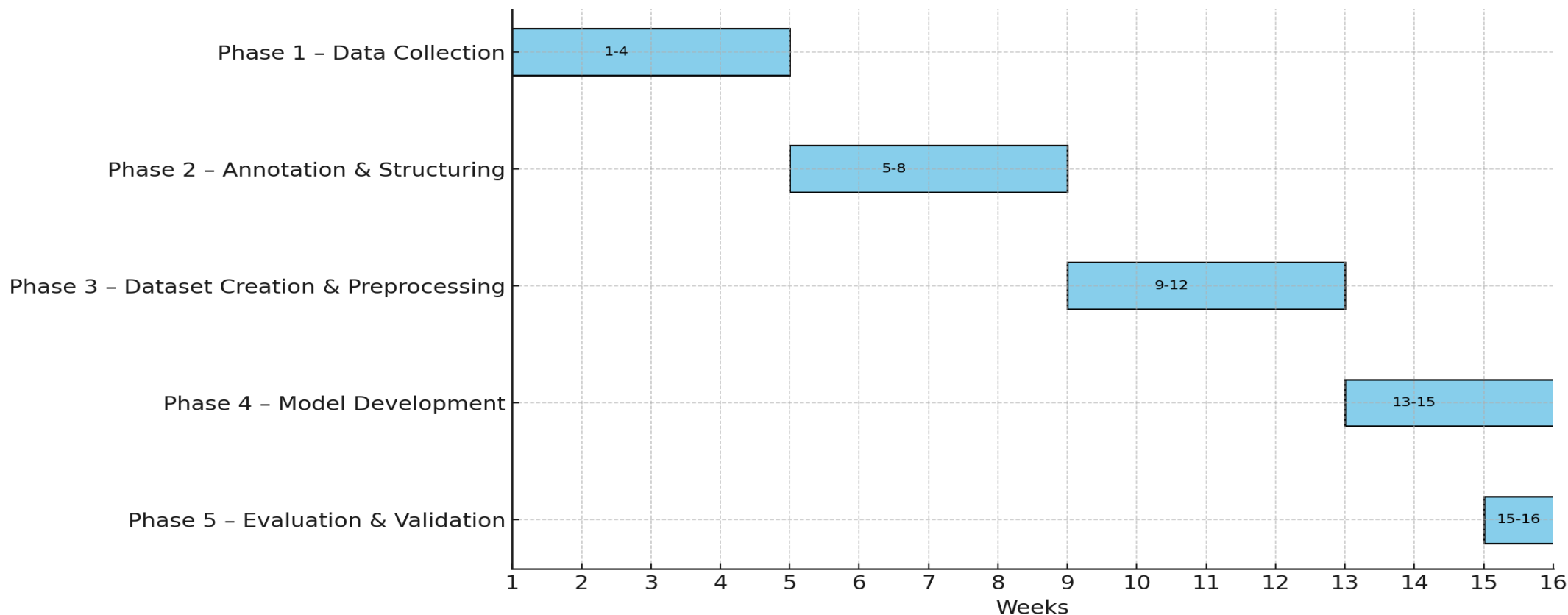


OVERVIEW OF IMPLEMENTATION



```
▼ "root" : { 5 items
  "term" : string "Earnings Per Share"
  "definition" :
    string "Net profit divided by total outstanding shares, indicating
    profitability on a per-share basis"
  "significance" :
    string "Primary indicator of company profitability and stock valuati
  "usage" : string "Fundamental analysis and investment screening"
  ▼ "implications" : { 2 items
    "increasing" :
      string "Improved company profitability and shareholder value"
    "decreasing" : string "Declining profitability or share dilution"
```

TASK IMPLEMENTATION



TEST CASE - 1

- ***My landlord is not returning my security deposit even after I vacated the house properly. What should I do?***

Category: *Tenancy Dispute / Security Deposit*

- **Relevant Laws/Sections:**

- *Indian Contract Act, Section 73 (Compensation for loss or damage caused by breach of contract)*
- *Transfer of Property Act, Section 108 (Rights and liabilities of lessor and lessee)*

- **Probable Violations:** *Unjust Enrichment / Breach of Rental Agreement*

- **Advice/Steps to Take:**

- **Send a legal notice** to the landlord demanding the return of your deposit.
- **Keep documentation** such as rent receipts, agreement copy, photos of the vacated property.
- **Approach the Rent Control Board** or file a case in **Small Causes Court** (if amount is below ₹10 lakhs in some states).
- You may represent yourself using the complaint draft and reference judgments provided.

- **Reference Case:**

- *Rajeev vs Suresh (Delhi HC, 2019)* – Tenant awarded full refund plus compensation for mental agony.

TEST CASE - 2

- ***My employer has not paid my salary for the last two months despite repeated requests. What should I do?***

Category: *Labour and Employment Law / Salary Not Paid*

- **Relevant Laws/Sections:**

- *Payment of Wages Act, 1936 – Ensures timely payment of wages to employees.*
- *Industrial Disputes Act, 1947 – Addresses disputes between employers and employees including wage-related grievances.*

- **Probable Violations:** *Non-payment of wages / Breach of employment contract*

- **Advice/Steps to Take:**

- **Send a written notice** to the employer formally requesting the payment of pending salary.
- **Keep documentation** such as salary slips, attendance records, emails/messages exchanged with the employer.
- **Approach the Labour Commissioner** or file a complaint with the appropriate labour court if the employer fails to respond.
- **Use grievance redressal mechanisms** provided by the company or state labour department.

- **Reference Case:**

- *Akshay S/O Rammilan Pal vs State Of Maharashtra Through Its Police Station (FIR No 727/2022)*

CONCLUSION

Pioneering Legal AI: We have developed a comprehensive framework for "LawBot," an ML-based chatbot designed to make legal assistance more accessible and affordable in India.

Bridging the Gap: Our model directly addresses the core problem of a lack of affordable and reliable legal guidance for a large section of society.

Robust Framework: The project integrates critical modules for data collection, preprocessing, model training, and a dynamic query engine, ensuring a robust and reliable system.

Future Impact: LawBot has the potential to empower citizens by providing them with foundational legal knowledge, enabling them to navigate common legal challenges with greater confidence and preparation.

REFERENCES

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- [3] I. Chalkidis, et al., “LawBERT: Pre-trained Language Model for Legal Text,” *arXiv preprint arXiv:2010.03816*, 2020.
- [4] P. Lewis, et al., “Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks,” *arXiv preprint arXiv:2005.11414*, 2020.
- [5] CiteCaseLAW, *Reference for the model's ability to extract and cite legal precedents*. (Placeholder, replace with a relevant citation extraction model or remove if unavailable.)
- [6] The Indian Contract Act, 1872; The Payment of Wages Act, 1936; The Transfer of Property Act, 1882. *Foundational legal documents of India*.



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THANK YOU